

JUNXIAO ZHANG

Department of Biological Systems Engineering, University of Nebraska-Lincoln

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EDUCATION

University of Nebraska–Lincoln

PhD. Biological/Biological Systems Engineering

Advisor: Dr. Yufeng Ge

Expected Dec 2026

Lincoln, Nebraska

University of Nebraska–Lincoln

M.S. Agricultural & Biological Systems Engineering

Advisor: Dr. Yufeng Ge

May 2023

Lincoln, Nebraska

The Ohio State University

B.S. Agricultural Engineering

May 2021

Columbus, Ohio

RESEARCH EXPERIENCE

Graduate Research Assistant

June 2021 - Present

University of Nebraska-Lincoln, Department of Biological System Engineering

- Conducted research in high-throughput plant phenotyping, focusing on innovative imaging techniques and AI-based data analysis
- Developed data pipelines for large-scale phenotypic analysis with AI applications

PUBLICATION

Peer-Reviewed Articles

1. **Zhang, J.**, Bai, G., Shi, Y., Awada, T., and Ge, Y. Monitoring crop stomatal traits by high-throughput plant phenotyping, a systematic review from leaf to canopy. *Plant Phenomics*. **Under Review**
2. Bai, G., **Zhang, J.**, Chamara, N., Scoby, D., Murad, O., Stoerger, V., Awada, T., and Ge, Y. Six years of innovation: A comprehensive review of cable-suspended field phenotyping system and its role in precision agriculture. *Journal of the ASABE*. **Under Revision**
3. **Zhang, J.**, Thapa, K., Bai, G. B., and Ge, Y. (2025). Improved estimation of stomatal conductance by combining high-throughput plant phenotyping data and weather variables through machine learning. *Agricultural Water Management*, 309:109321. <https://doi.org/10.1016/j.agwat.2025.109321>

Conference Papers

1. **Zhang, J.**, Thapa, K., Chamara, N., Bai, G., and Ge, Y. (2023). Estimating crop stomatal conductance from rgb, nir, and thermal infrared images. In Thomasson, J. A. and Bauer, C., editors, *Autonomous Air and Ground Sensing Systems for Agricultural Optimization and Phenotyping VIII*, volume 12539, page 125390A. SPIE. <https://doi.org/10.1117/12.2663888>

TEACHING EXPERIENCE

Co-Instructor

University of Nebraska-Lincoln

BSEN 460 Instrumentation & Controls

Autumn 2024 & Autumn 2025

- Delivered lectures and lab sessions, held office hours, and prepared experimental setups.
- Used project-based learning to help students learn through real-world challenges.

Teaching Assistant

University of Nebraska-Lincoln

BSEN 260 Instrumentation I

Spring 2024

BSEN 460 Instrumentation & Controls

Autumn 2022

- Delivered lab sessions, graded assignments, and prepared experimental setups

The Ohio State University

FABE 3130 Heat & Mass Transfer

Spring 2021

FABE 3150 System Dynamic & Electricity

Spring 2021

- Graded assignments, supported lab sessions, and prepared experimental setups

CONFERENCE PRESENTATION

1. **Zhang, J.**, Bai, G., Murad, M. O. F., & Ge, Y. Solar Zenith Angle Correction Improves Soybean Yield Prediction from Low-Altitude NDVI Measurements. CANVAS 2025, Salt Lake City, UT. (Oral)
2. **Zhang, J.**, Chamara, N., Bai, G., & Ge, Y. Estimate Stomatal Conductance of Maize and Soybean Plants in Greenhouse via Imaging and Pot Weighting. ICPA 2024, Manhattan, KS; also presented at ASABE 2024, Anaheim, CA. (Poster)
3. **Zhang, J.**, Chamara, N., Bai, G., & Ge, Y. Diurnal Variation of NDVI for Soybean and Maize under Different Water Treatments. NAPPN 2024, West Lafayette, IN. (Oral)
4. **Zhang, J.**, Chamara, N., Thapa, K., Bai, G., & Ge, Y. Estimating Crop Stomatal Conductance from RGB, NIR, and Thermal Infrared Images. SPIE 2023, Orlando, FL. (Oral)
5. Thapa, K., **Zhang, J.**, Bai, G., & Ge, Y. Characterization of Maize Responses to Differential Nitrogen Rates using Image-Based Phenotyping. NAPPN 2023, St. Louis, MO. (Oral)
6. **Zhang, J.**, Chamara, N., Thapa, K., Bai, G., & Ge, Y. Estimating Maize and Soybean Stomatal Conductance Based on Time Series Canopy Temperature, NDVI and Weather Conditions. NAPPN 2023, St. Louis, MO. (Oral)
7. **Zhang, J.**, Thapa, K., Bai, G., & Ge, Y. Estimating Winter Wheat Stomatal Conductance Using Thermal and Spectral Imaging, Weather Variables, and Machine Learning. ASABE 2022, Houston, TX. (Oral)

AWARDS & HONORS

College of Engineering Graduate Student Teaching Fellow

2025-2026

University of Nebraska-Lincoln

Dean's Fellowship

2026

University of Nebraska-Lincoln

First Prize, Student Oral Presentation Competition

2025

Airborne and Satellite Remote Sensing Community, American Society of Agronomy

College of Engineering Professional Development Fellowship

2025

University of Nebraska-Lincoln

Milton Mohr Fellowship

2024

University of Nebraska-Lincoln

David H. and Annie E. Larrick Graduate Student Travel Award

2022

University of Nebraska-Lincoln

Dean's List

2021

The Ohio State University

PROFESSIONAL DEVELOPMENTS

CIRTL Associate Certification

2026

Center for the Integration of Research, Teaching and Learning

PROFESSIONAL ACTIVITIES

Secretary

2023 - 2024

The Association of Overseas Chinese Agricultural, Biological, and Food Engineers

Website Editor

2022 - 2024

The Association of Overseas Chinese Agricultural, Biological, and Food Engineers

ACADEMIC SERVICE

Peer Reviewer

The Plant Phenome Journal

Undergraduate Proposal Reviewer

2023 - 2024

University of Nebraska-Lincoln

PROFESSIONAL SOCIETY MEMBERSHIPS

ASA, CSSA, and SSSA

Since 2025

The International Society of Precision Agriculture

Since 2024

North American Plant Phenotyping Network

Since 2022

American Society of Agricultural and Biological Engineers

Since 2021

SKILLS

Programming languages & Software

- Python, MATLAB, R, C/C++, Linux, LabVIEW, $\text{\LaTeX}$
- SOLIDWORKS, AutoCAD

Languages

- English (Proficient), Chinese (Native)